PASS Summit

AI Enabled Fabric in your Data Practice

Monday, November 9th, 2026

Presented by Ginger Grant

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Course Setup

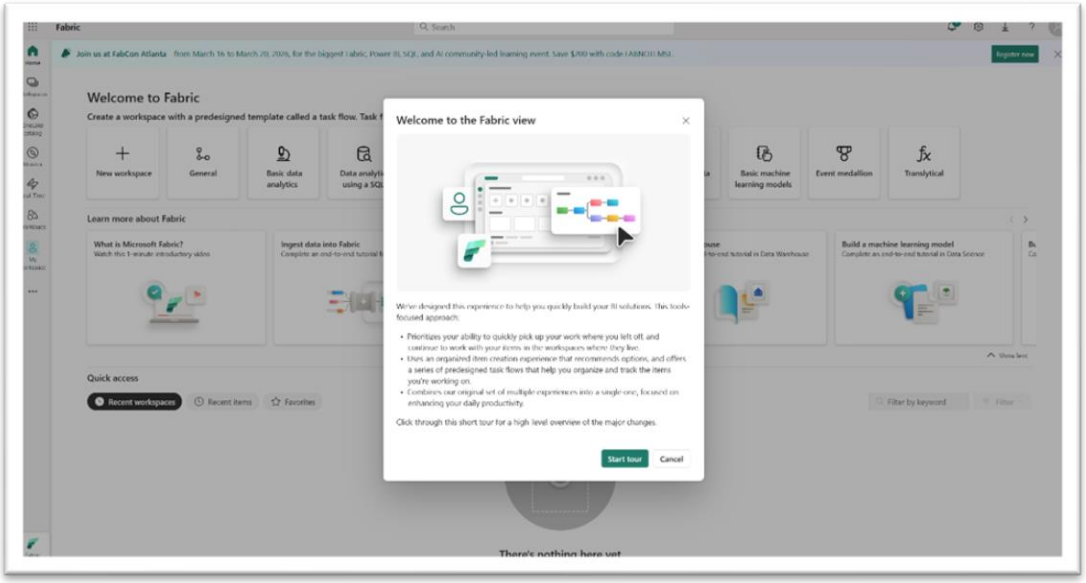
1. In your browser, go to https://app.fabric.microsoft.com/



2. Login with the email address **User#.Fabric@desertislesql.com**

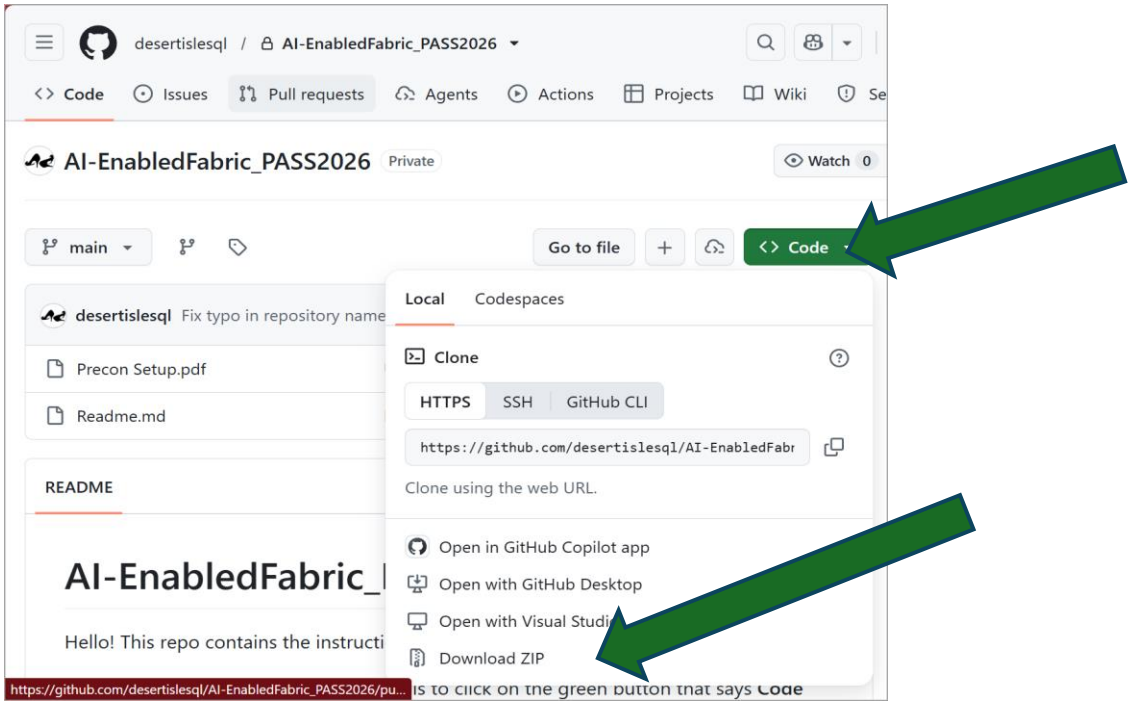
   Password is **tH3AI_P.55w0rd**

   # **-** use the assigned number



When you are successful you will see this screen. Congratulations! You are signed in and ready to work with Fabric

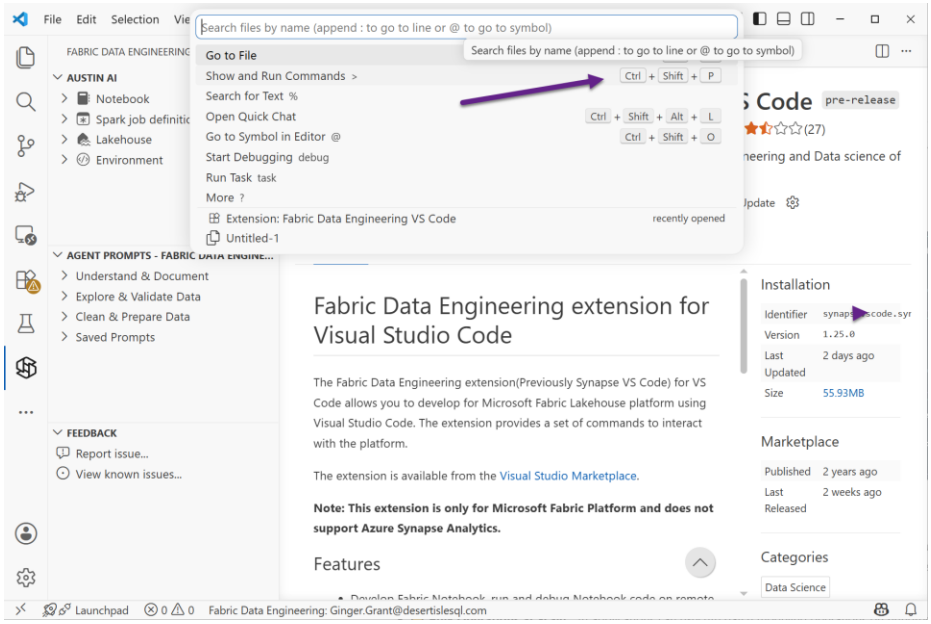
3. Download the contents of the github repo **https://github.com/desertislesql/AI-EnabledFabric_PASS2026**



4. Click on the green **Code** button when you do you will see a popup menu.
5. Select the option to **Download Zip** Remember where you save and unzip this as we will be using it in class
6. For class you will also need to use **SQL Server Management Studio 22**.
   Download SQL Server Management Studio 22 from https://learn.microsoft.com/en-us/ssms/install/install
7. Class exercises also include **VSCode**.
   Download it from https://code.visualstudio.com/

VS Code Fabric Data Engineering with GitHub Copilot

1. In your Browser, go to app.fabric.Microsoft.com
2. Go to the Workspace Enabling AI
3. Create a folder where you can store all the work for the class using either your name or your initials.
4. Open up VSCode and select File->New Window
5. In the new window, select the extension icon and check to see that you have Fabric Data Engineering installed
6. Click on the search bar at the top of the screen and Select **Ctrl+Shift+P**

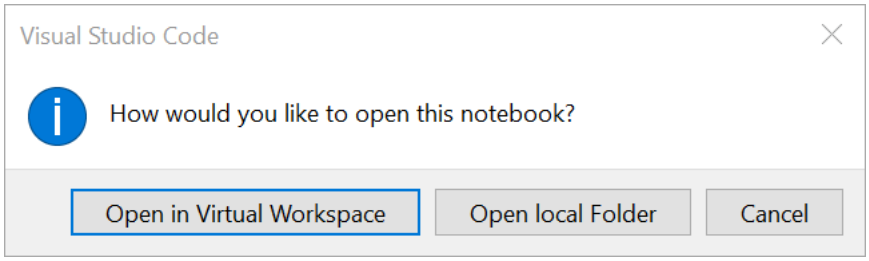


7. Select Fabric Data Engineering: Select Workspace
8. From the list select **Enabling AI**
9. Go to the view menu and select Chat to open the GitHub.
10. Modify the prompt to substitute your folder and initials Type the prompt In the Enabling AI workspace in the folder <your folder> Create a lakehouse called <yourInitials>_lkData
11. Go to the Fabric service and open up the lakehouse you created
12. Create a new table based on a shortcut from XXX_OpDataStore. In the lakehouse explorer, click on the tables folder and from the magical ellipse menu select **New shortcut**
13. Select Microsoft OneLake
14. Select XXX_OpDataStore lakehouse and click the **Next** button.
15. Access the table: dbo.Hawkeye_liquor
16. Click on the **Create** button to complete the shortcut

17. Go back to VSCode and add this new prompt substituting XXX with your initials

Create a new notebook in the workspace Enabling AI called XXX_nbData. Add two cells to the notebook. The first should be a configure statement to add XXX_lkData as the default lakehouse and create a select statement to get the categoryname, city and the total Saleamount from hawkeyeliquor

If you see this window



Select Open in Virtual workspace

18. Run the notebook in VSCode and then go back to the service and do the same.

End of Exercise

Fabric MCP Server

Make sure you install the Fabric MCP Server prior to completing this exercise. In VSCode.

1. In VSCode type Ctrl+Shift+P and select MCP: List Servers
2. If the Fabric MCP Server is not running start it
3. Go to theGitHub Copilot Pane making sure you are in Agent mode. Type this in the window

   What Fabric workload types are available?

   Review the results

4. Type this in the window
   List my Microsoft Fabric workspaces and find the workspace named Enabling AI
5. Inspect the Lakehouse
   In the Enabling AI workspace, inspect the lakehouse named XXX_lkData List the available tables, folders, and files.
   Ask for a cleaner summary:

Create a short inventory report for the XXX_lkData lakehouse showing tables, files, and folders.

6. Let's start documenting the Lakehouse
   Using the metadata from the XXX_lkData lakehouse, create documentation that includes:
   - Lakehouse purpose
   - List of tables
   - Probable business meaning of each table
   - Key columns
   - Potential data quality checks
   - Recommended owners or subject areas
   This will create the SQL Validation queries
7. Prompt copilot
   Create a Fabric notebook plan that profiles all tables in the XXX_OpDataStore lakehouse and writes the results to a data quality summary table

   Expected output will provide you with a plan containing:

   - Connect to the lakehouse.
   - Loop through lakehouse tables.
   - Capture row counts.
   - Capture null counts.
   - Capture duplicate counts.
   - Save results to a summary table.
   - Produce a final status report.

End of Exercise

Fabric CLI

For this demo we are going to migrate a notebook from Enabling AI test workspace to Enabling AI Test workspace.

1. Go to the command prompt by typing CMD in the Search window
2. Check to see if you have python installed in the path
   `python --version`
3. Install Fabric CLI

```
pip install ms-fabric-cli
```

4. Validate the install

```
Fab –version
```

5. Sign in to Fabric. When prompted, select Interactive with a web browser
6. Once you are logged in list all the items in the Enabling AI workspace

```
fab ls "Enabling AI" -l
```

7. Export the notebook definition.

```
fab export notebook `
    "Customer Sentiment Analysis" `
    --workspace "Enabling AI" `
    --path .\NotebookExport
```

8. Review the Notebook with AI by opening the notebook in VS Code. Add this prompt in GitHub Copilot:

Review this notebook and identify:
- Performance improvements
- Documentation gaps
- Security concerns
- Code quality issues

Review the results to see how you can improve the code.

9. Go back to the command prompt and type

```
fab ls "Enabling AI Test" -l
```

10. Deploy the Notebook and Import into the test workspace.

```
fab import notebook `
    --workspace "Enabling AI Test" `
    --path .\NotebookExport
```

11. Verify Deployment List notebooks in the target workspace.

```
fab ls "Enabling AI Test" -l
```

While you can do this in the UI, Fabric CLI provides another way to move notebooks. Instead of manually recreating notebooks in another workspace, Fabric CLI allows us to treat notebooks like code.

End of Exercise

Copilot Prompt Series: Build a Data Factory Pipeline

1. Select **New item**, then **Warehouse**.
2. Name it XXX_SalesWH where XXX are your initials and select **Create**.

In the workspace, select **New item > Pipeline** and name it PL_Northwind_Load.

On the **Home** tab, select **Copilot**.

Select the **Ingest data** starter. When a prompt needs a connection, type / to pick or create it.

Add this propt

```
Create a Copy activity named Copy_Orders that copies the Orders table
from the OData source
https://services.odata.org/V4/Northwind/Northwind.svc/ into a new
table named stg_Orders in the fabric warehouse SSX_Sales
```

3. In the folder of the git repo, open up the script called DWScript.sql in notepad
4. In your Browser, go to app.fabric.Microsoft.com
5. Go to the Workspace Enabling AI
6. Go to your folder which you created in a previous lab
7. Select **New item**, then **Warehouse**.
8. Name it XXX_SalesWH where XXX are your initials and select **Create**.
9. In SalesWH, in the ribbon at the top of the screen, select the icon for **New SQL query** and paste the warehouse script which you copied from the script called DWScript.sql
10. Select **Run**. The script will create a helper numbers table, three dimensions, one fact table with 2,000 sales lines covering 2024 and 2025, and non-enforced keys that help Power BI detect relationships.
11. Run this query and confirm the row counts.

```
SELECT 'DimCustomer' AS TableName, COUNT(*) AS RowCount FROM
dbo.DimCustomer

UNION ALL SELECT 'DimProduct', COUNT(*) FROM dbo.DimProduct

UNION ALL SELECT 'DimDate',    COUNT(*) FROM dbo.DimDate

UNION ALL SELECT 'FactSales',  COUNT(*) FROM dbo.FactSales;
```

You should see these results

Table	Expected rows
DimCustomer	20
DimProduct	12
DimDate	731
FactSales	2,000

2.1 Create the semantic model

1. In SalesWH, select **New semantic model** on the ribbon.
2. Name it SalesWH Model, confirm the workspace, and select the four tables: DimCustomer, DimProduct, DimDate, FactSales.
3. Select **Confirm**. The model is created in Direct Lake mode.

4. Go to the workspace, open SalesWH Model, and select **Open data model**.

2.2 Create or verify relationships

1. In the model view, check whether relationships were detected from the foreign keys.
2. If any are missing, select **Manage relationships > New relationship** and create each one:

From (many)	To (one)	Cardinality	Cross-filter
FactSales[CustomerKey]	DimCustomer[CustomerKey]	Many to one	Single
FactSales[ProductKey]	DimProduct[ProductKey]	Many to one	Single
FactSales[DateKey]	DimDate[DateKey]	Many to one	Single

3. Select **Assume referential integrity** on each relationship.

2.3 Add measures

Select the FactSales table, choose **New measure**, and create each of these. Set the format shown.

dax

Total Revenue = SUM (FactSales[Revenue])

Format: Currency, 0 decimals

dax

Total Quantity = SUM (FactSales[Quantity])

Format: Whole number

dax

Order Count = DISTINCTCOUNT (FactSales[OrderNumber])

Format: Whole number

dax

Average Order Value = DIVIDE ([Total Revenue], [Order Count])

Format: Currency, 2 decimals

2.4 Add metadata that helps Copilot

1. **Hide technical columns:** Hide CustomerKey, ProductKey, DateKey, SalesKey, and OrderNumber in all tables, and hide the FactSales[Revenue] column so Copilot uses the Total Revenue measure instead.
2. **Set data categories:** Set DimCustomer[Country] to **Country/Region** and DimCustomer[City] to **City** so map visuals work.
3. **Sort month names:** Select DimDate[MonthName] and set **Sort by column** to MonthNumber.
4. **Mark the date table:** Select DimDate, choose **Mark as date table**, and select the Date column.
5. **Add descriptions:** Add a description to Total Revenue: "Net revenue after discounts, in USD." Optionally, use Copilot to generate descriptions for the other measures and review them before accepting.

2.5 Prepare the model for AI

1. Return to the semantic model page in the workspace and select **Prep data for AI**.
2. In **Simplify the data schema**, clear any remaining key or helper fields, such as MonthNumber, and select **Apply**.
3. In **AI instructions**, enter:

Revenue always means the Total Revenue measure.

Top customers are ranked by Total Revenue, highest first, and show CustomerName.

Country comes from the DimCustomer table.

4. Select **Apply**, then **Close**.

Checkpoint: The model has three relationships, four formatted measures, hidden key columns, and AI instructions saved.

Part 3: Generate the report page with Copilot

3.1 Create a blank report

1. On the semantic model page, select **Create a report > Start from scratch**, or select **New report** from the data model ribbon.
2. If the report opens in Reading view, select **Edit**.

3.2 Prompt Copilot

1. On the ribbon, select **Copilot**.
2. Optional warm-up: enter Summarize the model and read how Copilot describes your tables and measures.
3. Enter the target prompt exactly:

Create a page showing revenue by country and top customers

4. Wait for Copilot to generate the page. Results vary between runs, but a typical page includes:
    - A bar chart or map of Total Revenue by Country
    - A bar chart or table of top customers by Total Revenue
    - One or more cards, such as Total Revenue or Order Count

3.3 Review the generated page

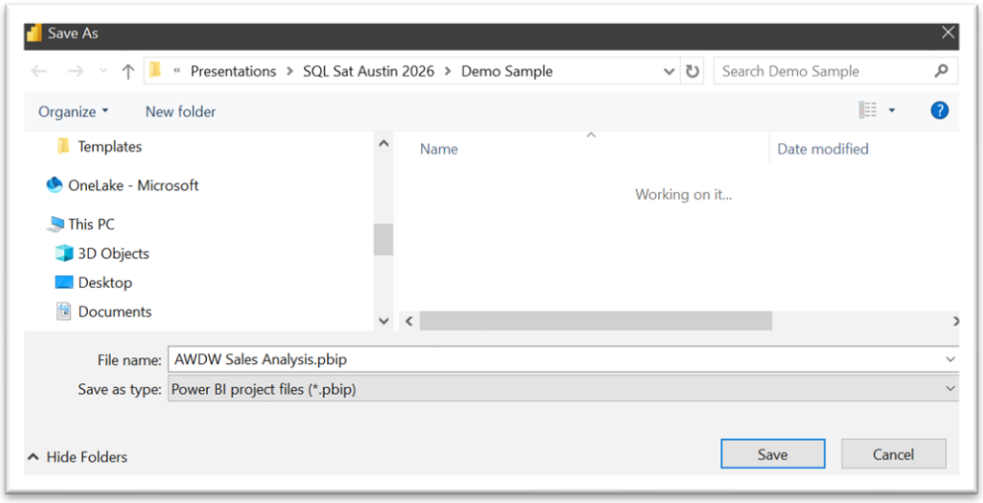
Answer these questions before changing anything:

1. Did Copilot use the Total Revenue measure or the hidden Revenue column?
2. Are countries and customers sorted by revenue, highest first?
3. Does the top customers visual limit the list, for example to the top 5 or 10?
4. Do the visual titles accurately describe what is shown?

Using GitHub Copilot with Power BI

1. Using the data downloaded from GitHubRepo, Open the PBIX file AWDW Sales Analysis.pbix
2. Go to the Model view by clicking on the model icon on the right side of the screen.
3. In the data view, click on the different tables and look at the description in the Properties Pane, look at all of the descriptions (spoiler alert, there aren't any)
4. Click on the x measures table and click on each of the DAX Measures.
5. In the data view, click on the different measure and look at the description in the Properties Pane for each DAX Measure, look at all of the descriptions (spoiler alert, there aren't any)

6. Go to File, Save as, browse to some location, and change the Save as Type to PBIP file.

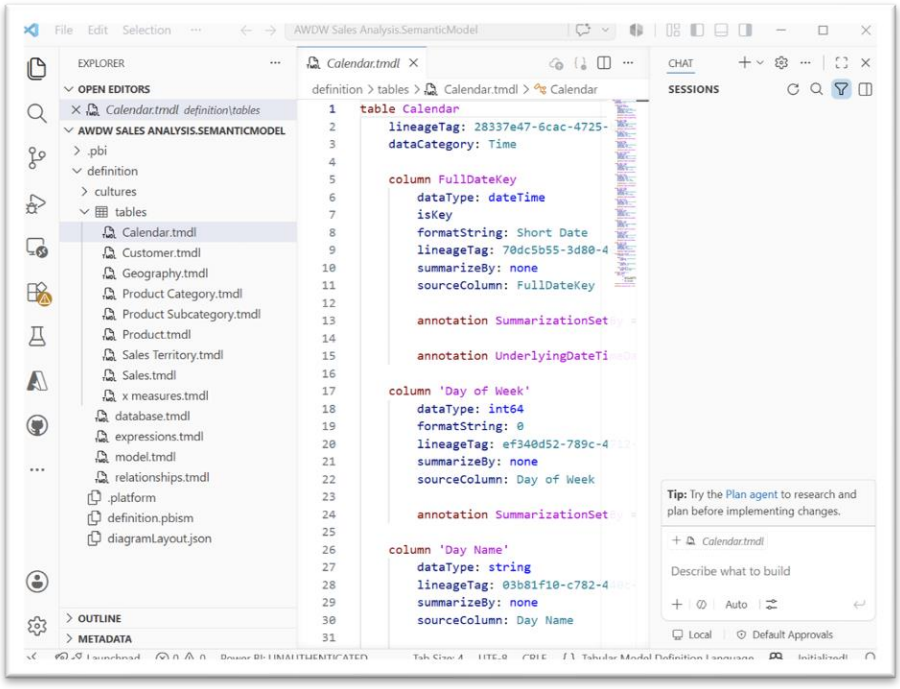


7. Click on the Save Button.

8. Close Power BI.

9. Open VSCode. Make sure you have the TMDL extension installed. Click on the extension and the folder AWDW Sales Analysis.SemanticModel.

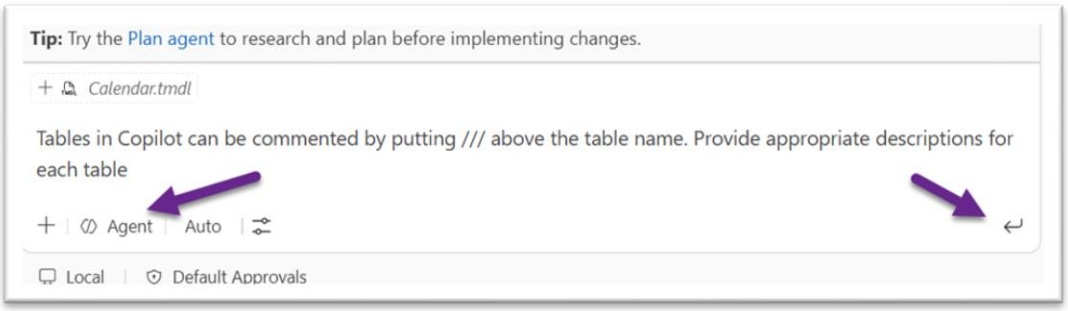
10. Open up the tables folder, then open up Calendar.tmdl. Your screen should look like the one below.



11. Open up the Copilot pane and enter this prompt

    Tables in Copilot can be commented by putting /// above the table name. Provide appropriate descriptions for each table

    Make sure that the agent mode is selected. It looks like this



12. Click on the send arrow

13. The copilot pane will fill up with things you should read. Tell it to keep the changes by clicking on the **Keep** button in the copilot pane if you want to click on it once. If you click on the keep button in the code pane you will be taken to each change which means you will have to click on it 8 times, one for each table.

14. Highlight the table x.Measures and enter the following prompt in the Github Copilot Chat

    Add descriptions for all of the DAX measures

    Click on the send arrow

15. Read all of the stuff in Github copilot and then click on the keep button.

16. Open up the report AWDW Sales Analysis.pbip in Power BI. Review changes and close.

End of Exercise

GINGER GRANT



You're collecting **gobs of data**.

Every department is **doing their thing**.

You're making **data-driven decisions**... inside the silos.

Isn't this enough? Not exactly.

TRUTH: Data is NOT driving you forward if you don't have a one system with an enterprise-level view: A single foundation that people understand, embrace and rally around. But it feels way too big to tackle.

You want to...

- → Implement new technology
- → Stay with the Microsoft Data Stack
- → Train your employees to use all of it.

But you may not know how. I understand and I can help.

THE OUTCOMES

- A custom architecture designed to scale
- Positioned for aligned investment in machine learning & AI
- Shared training & commitment to data quality
- Predictive analytics to help you plan
- Confident, data-driven decisions
- Data you can trust

ABOUT GINGER

Data Consultant since 2004

Microsoft Data Platform MVP since 2016, (384 worldwide)

Deep Understanding of Data Architecture | Vast Network of Resources

I Teach & Translate All Things Tech

I Code Custom Generative AI Solutions

MVP Microsoft® Most Valuable Professional

MCT

Let's Connect!

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